

[illegible]

Dataset	Macro F1	Non-cancer	Benign	Atypical	Malignant	MAE	RMSE	Spearman ρ	Pearson r
Balanced	0.517	0.7008	0.3803	0.3835	0.6035	0.002918	0.025611	-0.1394	-0.0051
Negative heavy	0.4433	0.6494	0.5825	0.4293	0.1119	0.001123	0.019196	0.1009	0.0131
Benign heavy	0.7249	0.8217	0.8555	0.4956	0.7266	0.001482	0.015133	-0.2361	-0.0074
Atypical heavy	0.3828	0.6657	0.2438	0.3234	0.298	0.002186	0.02577	0.0677	-0.0066
Malignant heavy	0.627	0.7884	0.4106	0.3797	0.9291	0.002402	0.018234	0.0832	-0.0086

The "benign-heavy" VMoE is the strongest individual model at 0.7249

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Attention regression Spearman ρ is near zero across all VMOEs, indicating the auxiliary regression task does not learn a meaningful ordering of CLAM attention scores.

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Class	Precision	Recall	F1	AUC-ROC	Support
non-cancer	0.6142	0.8157	0.7008	0.9208	445
benign	0.4782	0.3157	0.3803	0.5821	624
atypical	0.3078	0.5084	0.3835	0.7101	356
malignant	0.7189	0.52	0.6035	0.8309	600
macro avg	0.5298	0.54	0.517	—	2025

The balanced VMoE test macro F1 (0.5170) is substantially lower than its validation F1 (0.6763), indicating overfitting or distribution shift between splits.

	non-cancer	benign	atypical	malignant
non-cancer	363	32	44	6
benign	150	197	192	85
atypical	70	74	181	31
malignant	8	109	171	312

The balanced VMoE test macro F1 (0.5170) is substantially lower than its validation F1 (0.6763), indicating overfitting or distribution shift between splits.

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Class	Precision	Recall	F1	AUC-ROC	Support
non-cancer	0.874	0.5167	0.6494	0.9307	1289
benign	0.5592	0.6078	0.5825	0.8089	645
atypical	0.3204	0.6505	0.4293	0.6984	558
malignant	0.1829	0.0806	0.1119	0.4842	186
macro avg	0.4841	0.4639	0.4433	—	2678

	non-cancer	benign	atypical	malignant
non-cancer	666	148	470	5
benign	25	392	223	5
atypical	71	67	363	57
malignant	0	94	77	15

[illegible][illegible]

	non-cancer	benign	atypical	malignant	
non-cancer	401	3	44	34	benign_heavy is the strongest individual specialist on the test set (0.7249), achieving the highest benign F1 (0.8555) as expected from its training bias.
benign	1	687	131	81	
atypical	83	12	140	13	
malignant	9	4	2	190	

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[illegible]

Class	Precision	Recall	F1	AUC-ROC	Support			non-cancer	benign	atypical	malignant		
non-cancer	0.5097	0.9595	0.6657	0.9232	247		non-cancer	237	7	2	1		atypical_heavy has the lowest macro F1 (0.3828) on test — the
benign	0.1805	0.3755	0.2438	0.5025	237		benign	104	89	8	36		atypical class is visually the most ambiguous, and its precision (0.8447) with very low
atypical	0.8447	0.2	0.3234	0.6397	680		atypical	111	358	136	75		recall (0.2000) indicates the model is overly conservative, missing most atypical patches.
malignant	0.2533	0.3619	0.298	0.7748	105		malignant	13	39	15	38		
macro avg	0.4471	0.4742	0.3828	—	1269								

	non-cancer	benign	atypical	malignant	
non-cancer	237	7	2	1	atypical_heavy has the lowest macro F1 (0.3828) on test — the atypical class is visually the most ambiguous, and its precision (0.8447) with very low recall (0.2000) indicates the model is overly conservative, missing most atypical patches.
benign	104	89	8	36	
atypical	111	358	136	75	
malignant	13	39	15	38	

atypical_heavy has the lowest macro F1 (0.3828) on test — the atypical class is visually the most ambiguous, and its precision (0.8447) with very low recall (0.2000) indicates the model is overly conservative, missing most atypical patches.

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Class	Precision	Recall	F1	AUC-ROC	Support
non-cancer	0.8333	0.748	0.7884	0.961	254
benign	0.3553	0.4865	0.4106	0.7086	111
atypical	0.3383	0.4327	0.3797	0.7684	104
malignant	0.9658	0.895	0.9291	0.985	600
macro avg	0.6232	0.6406	0.627	—	1069

non-cancer benign atypical malignant

non-cancer benign atypical malignant

malignant_heavy achieves the highest malignant F1 (0.9291) and strong non-cancer recall, confirming specialisation toward the most visually distinct class.

	non-cancer	benign	atypical	malignant	
non-cancer	190	37	25	2	malignant_heavy achieves the highest malignant F1 (0.9291) and strong non-cancer recall, confirming specialisation toward the most visually distinct class.
benign	2	54	39	16	
atypical	34	24	45	1	
malignant	2	37	24	537	

malignant_heavy achieves the highest malignant F1 (0.9291) and strong non-cancer recall, confirming specialisation toward the most visually distinct class.

Phase 2											
Stacking Ensemble Meta-Learner											
Macro F1	Macro AUC-ROC	Matthews CC	Cohen's Kappa	MAE	RMSE	Spearman ρ	Pearson r				
0.8385	0.959	0.804	0.8001	0.016582	0.039244	-0.1162	-0.0183				
The ensemble achieves 0.8385 macro F1 on the held-out test set, a +32 pp gain over the best individual VMoE on the same balanced test split (balanced VMoE: 0.5170).											
Per-Class Classification Metrics						Confusion Matrix					
Class	Precision	Recall	F1	AUC-ROC		non-cancer	benign	atypical	malignant		
non-cancer	0.7708	0.9596	0.8549	0.9731	non-cancer	427	6	11	1		
benign	0.9431	0.7436	0.8315	0.9525	benign	50	464	74	36		
atypical	0.7178	0.736	0.7268	0.9211	atypical	73	15	262	6		
malignant	0.93	0.9517	0.9407	0.9893	malignant	4	7	18	571		
Cosine Similarity						Euclidean Distance					
	non-cancer	benign	atypical	malignant		non-cancer	benign	atypical	malignant		
non-cancer	1	0.4046	0.4722	0.1645	non-cancer	0	1.641	1.5977	2.1329		
benign	0.4046	1	0.5225	0.3838	benign	1.641	0	1.4825	1.7972		
atypical	0.4722	0.5225	1	0.2281	atypical	1.5977	1.4825	0	2.064		
malignant	0.1645	0.3838	0.2281	1	malignant	2.1329	1.7972	2.064	0		
Atypical remains the hardest class (F1 0.7268) with 20.5% of true atypical patches misclassified as non-cancer — the most common error, consistent with the visual ambiguity between atypical and normal epithelium.											
Non-cancer has the highest recall (96.0%) but the lowest precision (0.7708), meaning the ensemble is conservative: it prefers false non-cancer over false positives in the cancer classes.											
Malignant achieves the highest F1 (0.9407) and the largest Euclidean centroid separation from all other classes (min 2.06), confirming it is the most distinct class in the MetaLearner feature space.											
Benign and atypical centroids are the closest pair (cosine similarity 0.5225, Euclidean 1.4825), consistent with the known diagnostic difficulty distinguishing them											
The attention regression Spearman ρ (−0.1162) is near zero, confirming the auxiliary regression task does not transfer meaningful CLAM attention signal through distillation.											

No pseudo-labeling						Baseline					
VMoE Specialists											
Dataset	Macro F1	Benign	Atypical	Malignant		Dataset	Macro F1	Benign	Atypical	Malignant	
Balanced	0.5669	0.396	0.561	0.744		Balanced	0.517	0.3803	0.3835	0.6035	
Benign heavy	0.5573	0.548	0.645	0.479		Benign heavy	0.7249	0.8555	0.4956	0.7266	
Atypical heavy	0.3045	0.246	0.184	0.484		Atypical heavy	0.3828	0.2438	0.3234	0.298	
Malignant heavy	0.561	0.518	0.433	0.732		Malignant heavy	0.627	0.4106	0.3797	0.9291	
Stacking Ensemble											
Macro F1	Benign	Atypical	Malignant			Macro F1	Non-cancer	Benign	Atypical	Malignant	
0.4982	0.478	0.537	0.48			0.8385	0.8549	0.8315	0.7268	0.9407	
When training single model, without pseudo-labeling, in balanced dataset,											
the average macro F1 is slightly higher than the baseline (0.0499 respectively)											
But in other datasets, it perform worse than the baseline: 0.1676 (benign), 0.0783 (atypical), and 0.066 (malignant)											
The stacking ensemble, without pseudo-labeling, perform even worse than the balanced model. Which											
means the single models are overfitting on a noise, in this case, the unlabeled non-cancer patches.											
At the same time, the stacking ensemble performance decreased drastically compared to baseline (0.3624)											
Averaging Ensemble vs. Stacking Ensemble											
Model	Macro F1	Weighted F1	Macro AUC	MCC	Kappa	Per-Class F1					
Averaging Ensemble	0.8349	0.8463	0.9666	0.7957	0.7918	Class	Averaging	Stacking	Delta		
Stacking Ensemble	0.8385	0.8506	0.959	0.804	0.8001	non-cancer	0.8361	0.8549	0.019		
						benign	0.8354	0.8315	-0.004		
						atypical	0.7397	0.7268	0.013		
						malignant	0.9284	0.9407	-0.012		
The averaging ensemble (0.8349) nearly matches the stacking ensemble (0.8385) — a gap of only 0.4 macro F1 — despite having no learned parameters.											
Averaging achieves higher atypical F1 (0.7397 vs 0.7268, +1.3 pp) and higher macro AUC (0.9666 vs 0.9590), suggesting the MetaLearner does not improve on the											
hardest class and may slightly over-calibrate toward easier classes.											
The near-parity result supports the interpretation that the performance gain comes primarily from ensemble diversity (five biased VMoEs) rather than from the learned											
stacking weights. The MetaLearner adds marginal but consistent gains across most classes.											
Removing Regression Task											
VMoE						Baseline					
Dataset	Macro F1	Non-cancer	Benign	Atypical	Malignant	Dataset	Macro F1	Non-cancer	Benign	Atypical	Malignant
Balanced	0.5174	0.705	0.3711	0.3798	0.6136	Balanced	0.517	0.7008	0.3803	0.3835	0.6035
Negative heavy	0.4475	0.6736	0.5706	0.4289	0.1168	Negative heavy	0.4433	0.6494	0.5825	0.4293	0.1119
Benign heavy	0.688	0.8111	0.8283	0.5081	0.6043	Benign heavy	0.7249	0.8217	0.8555	0.4956	0.7266
Atypical heavy	0.3812	0.6775	0.1782	0.3565	0.3128	Atypical heavy	0.3828	0.6657	0.2438	0.3234	0.298
Malignant heavy	0.659	0.8456	0.469	0.3857	0.936	Malignant heavy	0.627	0.7884	0.4106	0.3797	0.9291
Ensemble											

Metric	Multi-Task (Baseline)	Single-Task (New)	Delta (Δ)	The simultaneous rise in Matthews Correlation Coefficient (MCC) and Cohen's Kappa is particularly significant; it indicates that the improvement isn't just a fluke in F1 calculation, but a genuine increase in the quality of the multi-class predictions.									
Macro F1	0.8385	0.8507	0.0122										
Macro AUC-ROC	0.959	0.9634	0.0044										
Matthews CC	0.804	0.8183	0.0143										
Cohen's Kappa	0.8001	0.8146	0.0145										
Class	Precision	Recall	F1	AUC-ROC	Class	Precision	Recall	F1	AUC-ROC				
non-cancer	0.7661	0.964	0.8537	0.9764	non-cancer	0.7708	0.9596	0.8549	0.9731				
benign	0.9426	0.7628	0.8432	0.9527	benign	0.9431	0.7436	0.8315	0.9525				
atypical	0.75	0.7584	0.7542	0.9312	atypical	0.7178	0.736	0.7268	0.9211				
malignant	0.9517	0.9517	0.9517	0.9934	malignant	0.93	0.9517	0.9407	0.9893				
The Single-Task model is clearly superior for your clinical objective. A 1.22% increase in Macro F1 combined with better Atypical detection is a high-value trade-off for a negligible 0.12% drop in the non-cancer F1. The model is now better at identifying the "hard" classes that matter most in subtyping.													
Raw attention score vs Attention percentile normalized													
VMoE					Baseline								
Dataset	Macro F1	Non-cancer	Benign	Atypical	Malignant	Dataset	Macro F1	Non-cancer	Benign	Atypical	Malignant		
Balanced	0.5357	0.7237	0.383	0.4086	0.6275	Balanced	0.517	0.7008	0.3803	0.3835	0.6035		
Negative heavy	0.8305	0.8095	0.8326	0.7134	0.9665	Negative heavy	0.4433	0.6494	0.5825	0.4293	0.1119		
Benign heavy	0.7727	0.7789	0.686	0.7019	0.9239	Benign heavy	0.7249	0.8217	0.8555	0.4956	0.7266		
Atypical heavy	0.7379	0.8394	0.7788	0.6432	0.6904	Atypical heavy	0.3828	0.6657	0.2438	0.3234	0.298		
Malignant heavy	0.6498	0.7085	0.6968	0.4786	0.7152	Malignant heavy	0.627	0.7884	0.4106	0.3797	0.9291		
The percentile-normalized approach shows significant increase across all single models, with the negative heavy model performance achieving close to Ensemble baseline													
Ensemble													
Metric	Baseline	Percentile normalized	Delta (Δ)										
Macro F1	0.8385	0.8398	0.0013										
Macro AUC-ROC	0.959	0.9642	0.0052	Class	Precision	Recall	F1	AUC-ROC					
Matthews CC	0.804	0.8048	0.0008	non-cancer	0.7708	0.9596	0.8549	0.9731					
Cohen's Kappa	0.8001	0.8012	0.0011	benign	0.9431	0.7436	0.8315	0.9525					
MAE	0.016582	0.063687	0.047105	atypical	0.7178	0.736	0.7268	0.9211					
RMSE	0.039244	0.117684	0.07844	malignant	0.93	0.9517	0.9407	0.9893					
Spearman ρ	-0.1162	0.8045	0.9207										
Pearson r	-0.0183	0.955	0.9733										
The percentile-normalized approach yields modest but consistent improvements across classification-oriented metrics													
Spearman's ρ increases from -0.1162 to 0.8045, and Pearson's r from -0.0183 to 0.955, reflecting a substantial shift from near-random or weak negative correlation to a very strong positive relationship. This indicates that the normalized outputs align much more closely with the underlying data distribution in terms of monotonicity and linearity.													